

Symmetries and Conservation HW-1

PROBLEM 1.

Piano string: In class, we have studied the longitudinal vibration of a stick. Here, we consider a much more complicated case. An elastic piano string can vibrate both transversely and longitudinally, and the two vibrations influence one another, as shown in Fig. 1. A Lagrangian that takes into account the lowest-order effect of stretching on the local string tension, and can therefore model this coupled motion, is

$$L[\xi, \eta] = \int dx \left\{ \frac{1}{2} \rho_0 \left[\left(\frac{\partial \xi}{\partial t} \right)^2 + \left(\frac{\partial \eta}{\partial t} \right)^2 \right] - \frac{\lambda}{2} \left[\frac{\tau_0}{\lambda} + \frac{\partial \xi}{\partial x} + \frac{1}{2} \left(\frac{\partial \eta}{\partial x} \right)^2 \right]^2 \right\}$$

Here $\xi(x, t)$ is the longitudinal displacement and $\eta(x, t)$ the transverse displacement of the string. Thus, the point that in the undisturbed string had coordinates $(x, 0)$ is moved to the point with coordinates $(x + \xi(x, t), \eta(x, t))$. The parameter τ_0 represents the tension in the undisturbed string, λ is the product of Young's modulus and the cross-sectional area of the string and ρ_0 is the mass per unit length.

1. (10 points) Use the least action principle to derive the two coupled equations of motion, one involving $\frac{\partial^2 \xi}{\partial t^2}$ and one involving $\frac{\partial^2 \eta}{\partial t^2}$.
2. (10 points) Show that when we linearize these two equations of motion, the longitudinal and transverse motions decouple. Find expressions for the longitudinal (c_L) and transverse (c_T) wave velocities in terms of τ_0 , ρ_0 , and λ .
3. (10 points) Assume that a given transverse pulse $\eta(x, t) = \eta_0(x - c_T t)$ propagates along the string. Show that this induces a concurrent longitudinal pulse of the form $\xi(x - c_T t)$. Show further that the longitudinal Newtonian momentum density in this concurrent pulse is given by

$$\rho_0 \frac{\partial \xi}{\partial t} = \frac{1}{2} \frac{c_L^2}{c_L^2 - c_T^2} T_1^0 \quad \text{where } T_1^0 = -\rho_0 \frac{\partial \eta}{\partial x} \frac{\partial \eta}{\partial t}$$

is the associated pseudo-momentum density.

SOLUTION:

- 1.

Consider the following variation $\xi, \eta \rightarrow \xi + \delta\xi, \eta + \delta\eta$ on the action:

$$S[\xi + \delta\xi, \eta + \delta\eta] - S[\xi, \eta] = \int_{\Omega} dt dx \left\{ \rho_0 \left(\frac{\partial\xi}{\partial t} \frac{\partial(\delta\xi)}{\partial t} + \frac{\partial\eta}{\partial t} \frac{\partial(\delta\eta)}{\partial t} \right) - \lambda \left[\frac{\tau_0}{\lambda} + \frac{\partial\xi}{\partial x} + \frac{1}{2} \left(\frac{\partial\eta}{\partial x} \right)^2 \right] \left(\frac{\partial(\delta\xi)}{\partial x} + \frac{\partial\eta}{\partial x} \frac{\partial(\delta\eta)}{\partial x} \right) \right\}$$

Integrate by part gives

$$\delta S = \int_{\Omega} dt dx \left\{ -\rho_0 \left(\frac{\partial^2 \xi}{\partial t^2} \delta\xi + \frac{\partial^2 \eta}{\partial t^2} \delta\eta \right) + \lambda \left[\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial\eta}{\partial x} \frac{\partial^2 \eta}{\partial x^2} \right] \delta\xi + \left[\lambda \left(\frac{\tau_0}{\lambda} + \frac{\partial\xi}{\partial x} + \frac{1}{2} \left(\frac{\partial\eta}{\partial x} \right)^2 \right) \frac{\partial^2 \eta}{\partial x^2} + \lambda \left(\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial\eta}{\partial x} \frac{\partial^2 \eta}{\partial x^2} \right) \frac{\partial\eta}{\partial x} \right] \delta\eta \right\} \\ + \int_{\partial\Omega} d\vec{S} \cdot \left\{ \rho_0 \left(\frac{\partial\xi}{\partial t} (\delta\xi) \vec{e}_t + \frac{\partial\eta}{\partial t} (\delta\eta) \vec{e}_t - \lambda \left[\frac{\tau_0}{\lambda} + \frac{\partial\xi}{\partial x} + \frac{1}{2} \left(\frac{\partial\eta}{\partial x} \right)^2 \right] ((\delta\xi) \vec{e}_x + \frac{\partial\eta}{\partial x} (\delta\eta) \vec{e}_x) \right\}$$

By boundary condition that $\delta\xi|_{\partial\Omega} = \delta\eta|_{\partial\Omega} \equiv 0$, and the arbitrariness of $\delta\xi, \delta\eta, \delta S = 0$ gives:

$$\begin{cases} -\rho_0 \frac{\partial^2 \xi}{\partial t^2} + \lambda \left[\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial\eta}{\partial x} \frac{\partial^2 \eta}{\partial x^2} \right] = 0 \\ -\rho_0 \frac{\partial^2 \eta}{\partial t^2} + \lambda \left(\frac{\tau_0}{\lambda} + \frac{\partial\xi}{\partial x} + \frac{1}{2} \left(\frac{\partial\eta}{\partial x} \right)^2 \right) \frac{\partial^2 \eta}{\partial x^2} + \lambda \left(\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial\eta}{\partial x} \frac{\partial^2 \eta}{\partial x^2} \right) \frac{\partial\eta}{\partial x} = 0 \end{cases}$$

and the equation of motion has been derived.

2.

When considering only the linear terms of the equations, we have:
$$\begin{cases} -\rho_0 \frac{\partial^2 \xi}{\partial t^2} + \lambda \frac{\partial^2 \xi}{\partial x^2} = 0 \\ -\rho_0 \frac{\partial^2 \eta}{\partial t^2} + \tau_0 \frac{\partial^2 \eta}{\partial x^2} = 0 \end{cases}$$

which implies the velocities are $c_L = \sqrt{\frac{\lambda}{\rho_0}}, c_T = \sqrt{\frac{\tau_0}{\rho_0}}$

3.

Substituting η in equation 1 with $\eta_0(x - c_T t)$ leads to

$$\rho_0 \left(c_L^2 \frac{\partial^2 \xi}{\partial x^2} - \frac{\partial^2 \xi}{\partial t^2} \right) = -\rho_0 c_L^2 \eta_0'(x - c_T t) \eta_0''(x - c_T t)$$

Consider the following transformation of coordinates: $\mu = x - c_T t, \nu = t$, then the equation reforms into $\rho_0 (c_L^2 \frac{\partial^2 \xi}{\partial \mu^2} + c_T^2 \frac{\partial^2 \xi}{\partial \mu^2} - \frac{\partial^2 \xi}{\partial \nu^2}) = -\rho_0 c_L^2 \eta_0'(\mu) \eta_0''(\mu)$. Suppose our boundary conditions are well enough to let $\rho_0 (c_L^2 + c_T^2) \frac{\partial^2}{\partial \mu^2}$ be self-adjoint operator (which has positive eigenvalues λ_n and corresponding eigenfunctions ϕ_n), then we have the eigenvalue decomposition:

$$\sum_n (\lambda_n c_n(\nu) - \rho_0 c_n''(\nu)) \phi_n(\mu) = \sum_n b_n \phi_n(\mu)$$

where b_n is the general Fourier coefficients of $-\rho_0 c_L^2 \eta_0'(\mu) \eta_0''(\mu)$, and c_n is the general Fourier coefficients of $\xi(\mu, \nu)$. It's easy to solve $c_n = \frac{b_n}{\lambda_n} + C_{n,1} e^{\sqrt{\frac{\lambda_n}{\rho_0}} \nu} + C_{n,2} e^{-\sqrt{\frac{\lambda_n}{\rho_0}} \nu}$. But to ensure the convergence along ν (which is time) and to maintain the finite energy, we must require both $C_{n,1}$ and $C_{n,2}$ to be zero. Thus $c_n = \text{Const}$ and $\xi(\mu, \nu) = \xi(\mu)$, which proves the first assertion.

To prove the second assertion, we firstly observe that $\frac{\partial^2 \xi}{\partial x^2} = \frac{1}{c_T^2} \frac{\partial^2 \xi}{\partial t^2}$, then

$$\rho_0 \frac{c_L^2 - c_T^2}{c_T^2} \frac{\partial^2 \xi}{\partial t^2} = -\rho_0 c_L^2 \eta_0'(x - c_T t) \eta_0''(x - c_T t)$$

In order to obtain $\rho_0 \frac{\partial \xi}{\partial t}$, integrate both side with t we have:

$$\rho_0 \frac{c_L^2 - c_T^2}{c_T^2} \frac{\partial \xi}{\partial t} = \frac{\rho_0 c_L^2}{2 c_T} (\eta_0'(x - c_T t))^2 = -\frac{\rho_0 c_L^2}{2 c_T^2} \left(\frac{\partial \eta}{\partial x} \right) \left(\frac{\partial \eta}{\partial t} \right)$$

Finally we get:

$$\rho_0 \frac{\partial \xi}{\partial t} = \frac{c_L^2}{2(c_L^2 - c_T^2)} \left(-\frac{\partial \eta}{\partial x} \frac{\partial \eta}{\partial t} \right)$$

PROBLEM 2.

1. (10 points) A second rank tensor F which is antisymmetric in one coordinate frame ($F_{\mu\nu} = -F_{\nu\mu}$) is anti-symmetric, show that
 - it is anti-symmetric in all frames.
 - the contravariant components are also antisymmetric $F^{\mu\nu} = -F^{\nu\mu}$ in this frame, and also antisymmetric in all frames.
2. (5 points) Show that any symmetric tensor $S_{\mu\nu} = S_{\nu\mu}$ has the same properties listed above.
3. (10 points) In an n -dimensional space, how many independent components are there for a generic r -rank tensor $T_{\alpha_1\alpha_2\cdots\alpha_r}$? How many independent components if the tensor is totally symmetric? What about if the tensor is totally anti-symmetric with (1) $r \leq n$, (2) $r > n$?
4. (10 points) In a four-dimensional space, define the anti-symmetric tensor $\epsilon_{\alpha\beta\gamma\sigma}$ with one of the components $\epsilon_{0123} = 1$
 - Prove that this tensor is invariant under Lorentz transformations.
 - Evaluate $\epsilon_{\alpha\beta\gamma\sigma}\epsilon^{\alpha\beta\gamma\sigma}$.
 - Show that for any rank-2 tensor A_β^α ,

$$\epsilon_{\alpha\beta\gamma\sigma} A_\mu^\alpha A_\nu^\beta A_\rho^\gamma A_\delta^\sigma = \epsilon_{\mu\nu\rho\delta} \det(A)$$

SOLUTION:

1.

Consider a coordinate transformation tensor Λ that acts on F , then $F'_{\rho\theta} = \Lambda_\rho^\mu \Lambda_\theta^\nu F_{\mu\nu}$. When swapping two indices, because it is evident that $\Lambda_\rho^\mu \Lambda_\theta^\nu = \Lambda_\theta^\mu \Lambda_\rho^\nu$, then $F'_{\theta\rho} = \Lambda_\theta^\mu \Lambda_\rho^\nu F_{\mu\nu} = \Lambda_\theta^\nu \Lambda_\rho^\mu F_{\nu\mu} = -\Lambda_\rho^\mu \Lambda_\theta^\nu F_{\mu\nu} = -F'_{\rho\theta}$. Thus F is anti-symmetric in all frames.

Suppose the metric tensor is $\eta_{\mu\nu}$ and that $(\eta^{-1})^{\mu\nu} = \eta^{\mu\nu}$ (so it is a standard form of Lorentz metric). Then $F^{\mu\nu} = \eta^{\mu\rho} \eta^{\nu\theta} F_{\rho\theta}$. When swapping two indices, because it is evident that $\eta^{\mu\rho} \eta^{\nu\theta} = \eta^{\nu\rho} \eta^{\mu\theta}$, then $F^{\nu\mu} = \eta^{\nu\rho} \eta^{\mu\theta} F_{\rho\theta} = \eta^{\mu\rho} \eta^{\nu\theta} F_{\theta\rho} = -\eta^{\mu\rho} \eta^{\nu\theta} F_{\rho\theta} = -F^{\mu\nu}$ is anti-symmetric. It is anti-symmetric in all frames can be proved by observing the covariant tensor $F_{\mu\nu}$ is anti-symmetric in all frames.

2.

By considering the procedure similar to 1, we have $S'_{\theta\rho} = \Lambda_\theta^\mu \Lambda_\rho^\nu S_{\mu\nu} = \Lambda_\theta^\nu \Lambda_\rho^\mu S_{\nu\mu} = S \Lambda_\rho^\mu \Lambda_\theta^\nu S_{\mu\nu} = S'_{\rho\theta}$. Thus S is symmetric in all frames, and $S^{\nu\mu} = \eta^{\nu\rho} \eta^{\mu\theta} S_{\rho\theta} = \eta^{\mu\rho} \eta^{\nu\theta} S_{\theta\rho} = \eta^{\mu\rho} \eta^{\nu\theta} S_{\rho\theta} = S^{\mu\nu}$ is symmetric.

3.

For a generic r -rank tensor, each index has n DOF, thus the independent components are in total n^r .

When T is symmetric, it is equivalent to the famous *Stars and Bars* problem (putting the same r balls into n different boxes), which has a well-known result of $\binom{n+r-1}{r}$.

When T is totally anti-symmetric, then all indices sequence that have repeated numbers will be corresponding to a trivial component of T . Thus

(1) when $r \leq n$, there are $\binom{n}{r}$ independent components. (Choosing r different indices)

(2) when $r > n$, the tensor is trivial by pigeon hole principle.

4.

We firstly solve the second and the third problem. $\epsilon_{\alpha\beta\gamma\sigma}\epsilon^{\alpha\beta\gamma\sigma}$ is zero for any indices sequence that has repeated indices, and is 1 for any permutation of 1234, thus $\epsilon_{\alpha\beta\gamma\sigma}\epsilon^{\alpha\beta\gamma\sigma} = \text{numbers of permutations of 1234} = 4! = 24$.

To solve the third problem, we remember that $\det(A) = \epsilon_{\alpha\beta\gamma\sigma}A_0^\alpha A_1^\beta A_2^\gamma A_3^\sigma$, and $\epsilon_{\mu\nu\rho\delta}\epsilon^{0123} = \epsilon_{\mu\nu\rho\delta}$. Thus

$$\epsilon_{\mu\nu\rho\delta}\det(A) = \epsilon_{\mu\nu\rho\delta}\epsilon_{\alpha\beta\gamma\sigma}A_0^\alpha A_1^\beta A_2^\gamma A_3^\sigma = \epsilon_{\mu\nu\rho\delta}\epsilon^{0123}\epsilon_{\alpha\beta\gamma\sigma}A_0^\alpha A_1^\beta A_2^\gamma A_3^\sigma = \epsilon_{\alpha\beta\gamma\sigma}A_\mu^\alpha A_\nu^\beta A_\rho^\gamma A_\delta^\sigma$$

Finally we are able to prove the first problem. Consider a Lorentz transformation Λ .

$$\epsilon'_{\mu\nu\rho\delta} = \epsilon_{\alpha\beta\gamma\sigma}\Lambda_\mu^\alpha \Lambda_\nu^\beta \Lambda_\rho^\gamma \Lambda_\delta^\sigma = \epsilon_{\mu\nu\rho\delta}\det(\Lambda) = \pm\epsilon_{\mu\nu\rho\delta}$$

If we consider only proper Lorentz transformation, that is $\Lambda \in SO(1, 3)$, then we have $\epsilon'_{\mu\nu\rho\delta} = \epsilon_{\mu\nu\rho\delta}$ and is Lorentz invariant.

PROBLEM 3. For any $n \times n$ matrix M , it can be proved that

$$\det(\exp(M)) = \exp(\text{Tr}(M))$$

1.(5 points) Prove Eq. 3 if M is a diagonal matrix.

2.(5 points) Prove Eq. 3 if M can be diagonalized by a similarity transformation, i.e. $M = UDU^{-1}$, where D is a diagonal matrix.

3.(5 points) Prove Eq. 3 for any matrix. (Hint: consider the Jordan canonical form.)

4.(5 points) Show that for a non-singular matrix H that depends on variable t , i.e. $H = H(t)$,

$$\frac{d}{dt}\det(H) = \det(H)\text{Tr}\left(H^{-1}\frac{d}{dt}H(t)\right)$$

SOLUTION:

1.

If $M = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a diagonal matrix, then $\exp(M) = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$, $\det(\exp(M)) = \prod_{i=1}^n e^{\lambda_i} = e^{\sum_{i=1}^n \lambda_i} = \exp(\text{Tr}(M))$.

2.

If M is diagonalizable, then $\exp(M) = \exp(UDU^{-1}) = \sum_{i=1}^{\infty} \frac{UD^nU^{-1}}{n!} = U \exp(D)U^{-1}$. Thus $\det(\exp(M)) = \det(\exp(D)) = e^{\sum_{i=1}^n \lambda_i}$ where $\{\lambda_i\}$ are the eigenvalues of M . It is well-known that the trace of a matrix is equal to the sum of all its eigenvalues, thus $\det(\exp(M)) = \exp(\text{Tr}(M))$.

3.

It is well-known that any matrix can be transformed (through a similar transformation) into block diagonal matrix $M = UDU^{-1}$, $D = \text{diag}(J_1, \dots, J_r)$, in which J_i is either diagonal (with some eigenvalue) or a Jordan block $J_{i,(j,j)} = \lambda$, $J_{i,(j,j+1)} = 1$.

Since we have proved the diagonal cases, it suffices to prove the Jordan block cases (because $\det(\exp(M)) = \exp(\text{Tr}(M))$ can be decomposed into $\prod \det(\exp(J_i)) = \exp(\sum \text{Tr}(J_i))$, and so we can treat each J_i separately). Suppose J is a Jordan block whose diagonal elements are all λ . Due to the fact that all upper-triangular matrices form a group under matrix multiplication (which means that upper-triangular matrices are closed under multiplication), it's easy to verify that J^n is a upper-triangular matrix with λ^n on its diagonal. Thus $\exp(J)$ is a upper-triangular matrix with e^λ on its diagonal. We all know that for upper-triangular matrices, $\det(\exp(J)) = \prod_{i=1}^n J_{(i,i)} = e^{n\lambda} = \exp(\text{Tr}(J))$, which completes the proof.

4.

$$\frac{d}{dt} \det(H) = \lim_{\Delta t \rightarrow 0} \frac{\det(H(t+\Delta t)) - \det(H(t))}{\Delta t}.$$

Because

$$H(t + \Delta t) = H(t)H^{-1}(t)H(t + \Delta t)$$

then

$$\det(H(t + \Delta t)) - \det(H(t)) = \det(H(t))(\det(H^{-1}(t)H(t + \Delta t)) - 1)$$

As a result of the continuity of the determinant function, it is reasonable to write the above equation as:

$$\det(H(t + \Delta t)) - \det(H(t)) = \det(H(t))(\det(I + H^{-1}(t)\frac{d}{dt}H(t)\Delta t) - 1 + o(\Delta t))$$

Notice that $I + H^{-1}(t)\frac{d}{dt}H(t)\Delta t \sim \exp(H^{-1}(t)\frac{d}{dt}H(t)\Delta t)$ as $\Delta t \rightarrow 0$, resulting in:

$$\frac{d}{dt} \det(H) = \lim_{\Delta t \rightarrow 0} \frac{\det(H(t))(\det(\exp(H^{-1}(t)\frac{d}{dt}H(t)\Delta t)) - 1)}{\Delta t}$$

By using the formula derived in 3, we obtain:

$$\frac{d}{dt} \det(H) = \det(H(t)) \lim_{\Delta t \rightarrow 0} \frac{\exp(\text{Tr}(H^{-1}(t) \frac{d}{dt} H(t) \Delta t)) - 1}{\Delta t}$$

Because the trace function is a linear function, $\text{Tr}(H^{-1}(t) \frac{d}{dt} H(t) \Delta t) = \Delta t \text{Tr}(H^{-1}(t) \frac{d}{dt} H(t))$, and we see immediately that

$$\lim_{\Delta t \rightarrow 0} \frac{\exp(\text{Tr}(H^{-1}(t) \frac{d}{dt} H(t) \Delta t)) - 1}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\exp(\Delta t \text{Tr}(H^{-1}(t) \frac{d}{dt} H(t))) - 1}{\Delta t} = \text{Tr}(H^{-1}(t) \frac{d}{dt} H(t))$$

Finally by substituting the limit back we have:

$$\frac{d}{dt} \det(H) = \det(H(t)) \text{Tr} \left(H^{-1} \frac{d}{dt} H(t) \right)$$

which completes the proof.

PROBLEM 4. For any square matrix M with complex elements, show that

- 1.(5 points) $MM^\dagger = M^\dagger M$, if M can be diagonalized by unitary transformation, i.e. $M = UDU^\dagger$, where $UU^\dagger = I$ and D is diagonal.
- 2.(10 points) if $M^\dagger M = MM^\dagger$, M can always be diagonalized by a unitary transformation.

SOLUTION:

1.

If $M = UDU^\dagger$, then $M^\dagger = (UDU^\dagger)^\dagger = UD^\dagger U^\dagger = M$. Thus $MM^\dagger = M^\dagger M$.

2.

To prove the statement, Shur's theorem is needed. It claims that for any matrix M , there exists a unitary transformation that $M = UTU^\dagger$ where T is an upper-triangular matrix.

Proof of Shur's Theorem: For any matrix $M_1 = M$, resulting from the fundamental principle of algebra, the polynomial $\det(M - \lambda I)$ has at least 1 root λ_1 , thus $\ker(M_1 - \lambda_1 I)$ is non-trivial. pick $\phi_1 \in \ker(M_1 - \lambda_1 I)$ so that $M_1 \phi_1 = \lambda_1 \phi_1$ (suppose ϕ_1 is normalized). Expand it to an orthonormal basis like $\phi_1, \psi_1, \dots, \psi_{n-1}$ ($\psi_1 \sim \psi_{n-1}$ don't have to be eigenvectors), and let $U = [\phi_1 \ \psi_1 \ \dots \ \psi_{n-1}]^\dagger$, then $T_1 = UM_1U^\dagger$ and the first column of T_1 is $T_{1,(1,1)} = \lambda_1, T_{1,(1,i)} = 0 (i > 1)$.

Recursively doing this step by projecting the matrix: $M_{i+1} = P^\dagger M_i P, P = [\psi_1 \ \dots \ \psi_{n-i}]$. And then find an eigenvalue of M_{i+1} and so on. It is apparent that the final matrix T_n is upper-triangular, which completes the proof.

Since

$$MM^\dagger = M^\dagger M \implies (UTU^\dagger)(UT^\dagger U^\dagger) = UTT^\dagger U^\dagger = (UT^\dagger U^\dagger)(UTU^\dagger) = UT^\dagger T U^\dagger$$

We have $TT^\dagger = T^\dagger T$. But because T is upper-triangular, by considering its trace we get that T is diagonal. Thus M is diagonalizable.